

1 Escalation Configuration

Overview

Menu	Master data → Escalation management → Escalation configuration
Transaction code	esccfg
Function authorization	esccfg

Purpose

Use this function to specify rules for the conversion of events into escalations (events provided by the applications). In the escalation configuration, you define the events that are converted to escalations, the recipients of specified escalations and how the escalation is technically sent.

Integration

To convert events into escalations, the escalation management uses the data stored in the escalation configuration.

Requirements

You have defined [Function groups](#) in the system if you:

- want to send an escalation to several recipients at the same time.
- want to send an escalation to specific persons of a group subject to defined factors, e.g. attendance.

You have linked HYDRA users of the [User administration](#) to persons of the [HR master data](#) to be able to send messages to these users/persons.

Field descriptions

Tab *Event*

The *Event* tab includes general information on an escalation. You can maintain the following information:

Event

This event triggers an escalation. Applications generate the event if specific situations occur.

Using the search function, you can select an event from the list of available events. You cannot enter events that are not available/included in the list.

Designation

You can enter a description of the event in this field. The system automatically suggests the description stored with the event when you select an event. You can overwrite this suggestion.

The name entered in this field is later used in the overview dialogs of current escalations. To easily identify the event, it is best to store a description that refers to the event and to the formula.

Recipient type

You can define the type of the recipient of a message in this field. The following types of recipients are available:

Function	You can select a function group that is notified.
Person	You can enter a personnel number for this recipient type.
Supervisor	The notification is sent to the supervisor of the person included in the escalation data. The acronym "PNR" identifies the person.
Automatic	The notification is sent to the person included in the escalation data. The acronym "PNR" identifies the person.
Workflow management	The escalation triggers a specified workflow.

Recipient

Depending on the selected type of recipient, you can enter the recipient here.

If you have selected "Person" as recipient type, you can use the search function to select the person from the HR master data.

If you have selected "Function" as recipient type, you can use the search function to select the recipient from the stored function groups.

Priority

You can use the priority in the message overview to sort the messages. The priority does not have any other function.

Farbe

Currently not used.

Active

This option specifies if the event configuration is enabled or not.

Cyclic monitoring

If the selected event is cyclically monitored, you must maintain the monitoring cycle here. The cycle is defined in days, hours and minutes. The default values are preallocated when you generate the monitoring. You can overwrite these values.

The system provides the event within the time specified as cycle. The conditions that might be stored for the escalation specify, if an event triggers an escalation. Example:

You have specified a cycle of 15 minutes for the cyclic escalation ABC.XYZ. The system therefore identifies the required data in the system every 15 minutes and provides an event. If the conditions stored in the escalation are fulfilled, the system converts the event into an escalation.



The escalation agent updates the internal list of configured events with the cyclic events stored in the database once an hour.

Automatic: after reading the message

Currently not used.

Automatic: after closing the event

Currently not used.

Automatic: after sending the message

If the event includes an automatic mode, you can view, enable and if required reconfigure the automatic here – depending on the definition of the event. You can define an automatic function that is performed when the message is sent (in case of an escalation).

The following automatic functions are available (depending on the event):

Function Close:

This function closes the event. You can configure that for example the escalation is automatically closed once the message has been sent.

If the function "Close after sending the message" is enabled, the message is sent and the event is closed with the comment "-".

Find below an example for the generation of a template:

You want to configure an escalation for the event TNR.OFFLINE for the terminals 11-14 that often cause problems. You want to forward this event to the system administration for a quick troubleshooting. Make the following entries in the *General* tab:

Field	Entry
Event	TNR.OFFLINE
Description	Terminal 11 - 14 OFFLINE
Recipient type	Function group
Recipient	Maintenance
Cyclic monitoring	0 hours, 00:30 minutes (the terminal status is checked every 30

minutes).

Workflow management

Workflow

Select the workflow that is triggered in case of this event.

User

HYDRA user that is transferred with the triggered workflow as user. Depending on the workflow, the defined user is also used in the workflow.

Group

Workflow group that is transferred with the triggered workflow as group. Depending on the workflow, the defined group is also used in the workflow.

Function

Workflow functions that are transferred with the triggered workflow as function. Depending on the workflow, the defined function is also used in the workflow.

Tab Options

Cyclic monitoring

This option defines if the event is generated via cyclic monitoring.

ESK mode

Type of escalation processing (currently only "F").

Automatic: after reading the message

Currently not used.

Automatic: after closing the event

Currently not used.

Automatic: after sending the message

See tab *Event*.

Tab Message

In the *Message* tab, you can maintain templates for the messages to be sent. Here, you enter the required message text and, if the selected type of notification is e-mail, also the subject.

Each event provides additional detail data. The detail data that an event provides is defined in the system. When you generate the template for a message, you can use placeholders for the provided detail data. If an event triggers an escalation, the placeholders in the template are replaced by the current values of the event and the complete current message text is generated.

You can maintain the following fields:

Subject

The text entered here is used as subject if the notification type e-mail has been selected. Also in the subject line, you can use placeholders that are replaced by the current values later on (see also message).

The system automatically uses the event name stored in the system as subject when the event is created. You can overwrite it.

Text

In the field *Text*, you edit the template for the actual message. You can store up to 320 characters for the message text. Note:

The maximum length depends on the message type. The following lengths are currently available:

E-mail: 320 characters

Text message (SMS): 320 characters (2 x 160 characters)

Pager: only the subject is transferred

The defined template might increase in size when the placeholders are replaced. When generating the message text, it might happen that the resulting text exceeds 320 characters. In this case, the placeholders cannot be replaced. With e-mail as communication type, the complete message text can be transferred.

In the template, you can store placeholders that are replaced by current values later on. The list of available placeholders shows the available placeholders.

Available placeholders

You can include placeholders in the message text, if you enter the identifier listed in the *Available placeholders*. Put a "%" before and after the identifier. Double-click the list of available placeholders to insert the selected placeholder at the end of the text template.

The message "Terminal 11 is offline" is generated from the template "Terminal %TNR.TNR% is offline" if this event occurs on terminal 11.

In the following, the example of the previous section is refined. In the *General* tab, you have already edited data for the event TNR.OFFLINE for the terminals 11-14. You want to notify the system administrators via e-mail and via text message (SMS). You can use a maximum of 2x160 characters (max. message length). The list of available placeholders displays the following detail data that is available for the event TNR.OFFLINE:

TNR.BEZX → Terminal location

TNR.BEZZ → Terminal designation/name

TNR.TNR → Terminal number

TNR.ZYKL:I → Actual cycle of status messages

TNR.ZYKL:S → Target cycle of status messages

System variables

In addition, the placeholders DAT (system date) and ZEI (system time) are available. They include date and time of message receipt in the server and can be formatted as follows:

%ZEI time hh:mm:ss% or

%ZEI time hh:mm%

or

%DAT date dd.mm.yyyy% or

%DAT date mm/dd/yyyy% or

%DAT date yyyy-mm-dd%

The placeholder WTG (weekday) includes the weekday of the system date. The placeholder is replaced by 3 characters describing the English weekdays: MON, TUE, WED, THU, FRI, SAT and SUN.

The placeholder BEM (comment on the escalation) is available for specific escalations. The escalation descriptions specify whether and how this parameter is assigned

The system provides the placeholders MSGPRIO, MSGCLASS and MSGRCV for specific escalations. The system collects the values priority/urgency (MSGPRIO), information class/importance (MSGCLASS) and recipient/addressee (MSGRCV) for a posting (e.g. reset maintenance in the terminal). These values are then transferred in the escalation (e.g. escalation RES.MAINTENANCE_RESET). You can refer to these values in the condition and in the message.

- The placeholder MSGPRIO includes the priority/urgency of a posting and can have the following values: empty (not specified), 1 (highest priority), 2 (high priority), 3 (normal priority), 4 (low priority) and 5 (lowest priority). If the message is sent via e-mail, the priority is taken over as mail priority.
- The placeholder MSGCLASS can include the following values: empty (not specified), I (information), W (warning) or E (error).
- The placeholder MSGRCV includes the group of addressees (not function group!), e.g. "plant manager".

Make the following entries in the *Message* tab:

Field	Entry
Subject	Terminal %TNR.TNR% OFFLINE

Text Terminal %TNR.TNR% at location %TNR.BEZK% has been offline at least since %DAT date dd.mm.yyyy%, %ZEI time hh:mm:ss%. Please check network availability.

Tab Notification

The *Notification* tab defines how the person or function group stored in the *Event* tab is notified. You can configure different notification combinations using logic combinations.

The notification types defined in this tab apply to all persons of a function group. Here, the system does not check if the required communication data is maintained in the HR master data for individual persons or for the member of a function group and if the data is maintained in the basic settings.

Currently the following notification types are available: Notification to the client (always possible) and notification via e-mail, SMS/text message and pager (if licensed). You can always configure these notification types – independent of licensing. The notification itself is only performed with the respective license.

You edit the notification type as follows:

Operator

The operator specifies how the notification types are combined with each other. You can use the following operators:

"TO"

The message is sent using the message type stored below.

"AND"

The message is additionally sent using this communication type.

"OR"

For the notification type of the higher level, the system checks if the notification can be realized successfully (it is checked if all data required for the notification is available). If this is not the case, the notification is sent using the notification type of this level.

You can therefore use different individual communication types within a function group.

The notification type is maintained top down with descending priority. For technical reasons, the system uses different methods to check the success of the individual notification types:

- Notification on the client

You can immediately check if the notification has been successful. If the person is currently logged on, the person is notified and the message is immediately displayed in the current message window.

- Notification via e-mail

Promptly after generation, the e-mail is transferred to the local SMTP server (mail server). If the transfer could be performed, the notification is identified as successfully transmitted.

If the e-mail delivery fails (e.g. due to a wrong e-mail address or because the server is not available), the escalation management does not correct the delivery success (e.g. by sending another message via a different communication channel).

- Notification via SMS/text message

Promptly after generation, the SMS/text message is sent using the licensed transmission process. The message is identified as successfully transferred if all data required for the transmission (phone number, directories, etc.) is maintained.

If the text message delivery fails (e.g. due to a wrong phone number or because the server/gateway is not available), the escalation management does not correct the delivery success (e.g. by sending another message via a different communication channel).

- Notification via (PZE) terminal

With this notification type it is configured that the message is displayed only once on the PZE terminal and it is removed after a maximum of 4 weeks. However, the message can be displayed several times during the period of the cyclic loading of authorizations. The message is displayed with the following actions on the PZE terminal: In, out, break, info, message and absence reason.

Messages can only be displayed on PZE terminals of type CT-36x, CT-37x and CT-38x. For more information, refer to the document "[Configuration Absence Workflow](#)".

- Notification via ERP system

The SAP Alert Management provides functionalities within the ECC to actively inform users of events that have occurred. If you additionally use the SAP Enterprise Portal, you can view and edit these alerts according to the requirements of the user.

If the escalation management is integrated into the SAP Alert Management and the SAP Enterprise Portal, you can transfer the generated escalations into the SAP Alert Management. The escalations are then available for the users of the SAP Enterprise Portal. This option requires the license ESK-SAPALM. For further information, please refer to the documentation ESK-SAPALM.pdf.

Send e-mail in cc to

Here, you can store an e-mail address that receives a copy of the sent e-mail.

The system only sends this copy, if you have configured the notification via e-mail. If you have exclusively configured other types of notification, an e-mail copy is not sent. The stored cc e-mail address is not used.

We use the example of the previous sections and configure below how the system administration is notified. The system administration does not work all the time, but the console agent is installed in the workplace PCs. But as the employees are not always present at their workplace, a second notification is sent via SMS/text message.

Make the following entries in the *Notification* tab:

Field	Entry	
	Operator	Notification type
Level 1	TO	Console
Level 2	AND	SMS/text message

Tab Condition

In the *Condition* tab, you define the conditions that trigger an escalation for an event. With each event, the escalation manager checks the conditions that are stored here. Only if the conditions are fulfilled and the result of the check is true, the escalation is generated or triggered.

You define the conditions using formulas and logical expressions. You link the expressions using comparison operators. You can combine and nest expressions using parenthesis .

In the formulas, you can include identifiers that can be used for conditions in the system. Find below the available identifiers for the conditions that the application transfers with the event.

You can use the following operators and expressions to formulate conditions:

"and"

Use the expression "and" to formulate AND constructions. An AND construction has the following structure:

< condition 1> and <condition 2 >

This condition is only true, if BOTH conditions are fulfilled.

This means: The condition "it rains and storms" is only true if it rains and storms at the same time. If it only rains and does not storm (or vice versa), the condition is not fulfilled.

"or"

Use the expression "or" to formulate OR constructions. An OR construction has the following structure:

< condition 1> or <condition 2 >

This condition is only true, if at least one of the two conditions is fulfilled.

The expression "it rains OR storms" is true if it only rains or only storms. If both expressions are true – it rains AND storms – the condition is also true.

"like"

Use the operator "like" to compare characters. You can use the operator as follows:

< value 1> like < example value 1 >.

For the example value, you can replace leading or trailing characters by additional placeholders:

"*" 0 –n any characters

"?" exactly 1 character (any)

The comparison "name like SCHMID*" is true for SCHMID and for SCHMIDT and also for SCHMIDLER. But it is not true for SCHMITT. If the comparison were "name like SCHMI*", it would be true for all names beginning with SCHMI, i.e. SCHMID, SCHMIDT, SCHMIDLER and SCHMITT.

"=="

If you use the operator "==" the condition must be fulfilled 100 %. You can use the operator to compare characters and to compare numbers. Use it as follows:

< value 1 > == <example value 1>

For the example mentioned above, the condition "name == "SCHMIDT"" is only true for SCHMIDT.

"!="

If you use the operator "!=" (not equal), the condition must not be fulfilled. You can use the operator to compare characters and to compare numbers. Use it as follows:

< value 1 > != <example value 1>

For the example mentioned above, the condition "name != "SCHMIDT"" is not true for SCHMIDT, but for all other names.

"<", ">", "<=", ">="

You can use the operators "<", ">","<=" , ">=" to compare numbers (and characters). Use them as follows:

< value > ">" <value 2>

The operators have the following meaning:

"<" less than...

">" greater than...

"<=" less than or equal to....

">=" greater than or equal to...

Make the following settings to store a condition:

Condition

Maintain the logical condition in the text field of the condition. You can use the above mentioned expressions, operators and combine them using parenthesis. You may only use identifiers that are provided by the event. The section "Variables" displays the available identifiers. If you use the identifiers in the condition, you must EXACTLY take over the syntax presented in the variables section.

A link at the end of this section includes an overview of functions, operators and constants that can be used in formulas and conditions.

If you use decimal numbers in formulas, use the dot "." as decimal separator.

Variables

This field shows the identifiers that are available for the respective event. In this field, you can also enter values for each identifier. You can store values that are expected in the application. Using these values, you can then immediately check if the formula is correct.

To check the result of the formula, the collected value for each identifier is entered in the formula and the formula is calculated. The formula returns the value "TRUE" or "FALSE" as a result. If the return value is "FALSE", you must either check the formula or the values. You must use ">"<" to begin and end strings, e.g. descriptions or words.

If you double-click a possible variable, this variable is inserted at the end of the formula.

System variables

System variables are provided in the same way as the variables.

In the previous chapters, it was discussed how to configure the notification of the system administration, if terminal 11-14 are offline. To complete this example, the formula is now stored.

The system administration should be notified, if at least one of the terminals 11 to 14 is affected. The event TNR.OFFLINE provides the identifier TNR.TNR (terminal number). The basic condition must be:

TNR.TNR >= 11

Here, the system would always trigger escalations if the terminal number were greater than or equal to 11. But you only want to generate messages, if the terminals 11-14 are affected. The condition is therefore extended.

TNR.TNR >= 11 and TNR.TNR <= 14

The condition is only true, if the terminal number is greater than or equal to 11 on the one side and on the other side it is less than or equal to 14.



Find a complete list of the supported operators [here](#).

Tab Condition – Examples for conditions

If you define conditions, you must make a difference between the data types CHAR (for characters) and INTEGER for numeric values or periods (of time).

If you compare CHAR data types, you must always enclose these CHAR data types by double quotes. You can compare numeric values without double quotes.

To find out if an acronym/field is numeric or if it is a string, you can check how the field is maintained in the system. If you can enter letters into the respective field, you must enclose the value by double quotes.

Example:	Condition
Machine status change to status 10 (MST.MALFUNCTION_OCCURRED) at a specific machine (alphanumeric) should trigger an escalation.	MNR.MNR == "WP1" and MST.MST == 10
Machine status change to status 10 (MST.MALFUNCTION_OCCURRED) at a specific machine (numeric) should trigger an escalation.	MNR.MNR == "123" and MST.MST == 10
A machine status 6 that exists longer than 10 minutes (MST.MALFUNCTION_CONTINUE) should trigger an escalation.	MST.MST == 6 and MST.DAUER >= 600
A machine status 6 that exists longer than 10 minutes at a specific machine (MST.MALFUNCTION_CONTINUE) should trigger an escalation, but only at a specific machine	MST.MST == 6 and MST.DAUER >= 600 and MNR.MNR = "1234"

(numeric).	
A machine status 6 that exists longer than 10 minutes at a specific machine (MST.MALFUNCTION_CONTINUE) should trigger an escalation, but only at a specific machine (alphanumeric).	MST.MST == 6 and MST.DAUER >= 600 and MNR.MNR = "WP1"
You want to check if less than 90 % of the target quantity (ANR.SGR:GUTP) of an operation is posted (ANR.EGR:GUT is the actual quantity)	ANR.EGR:GUT < (0.9*ANR.SGR:GUTP)

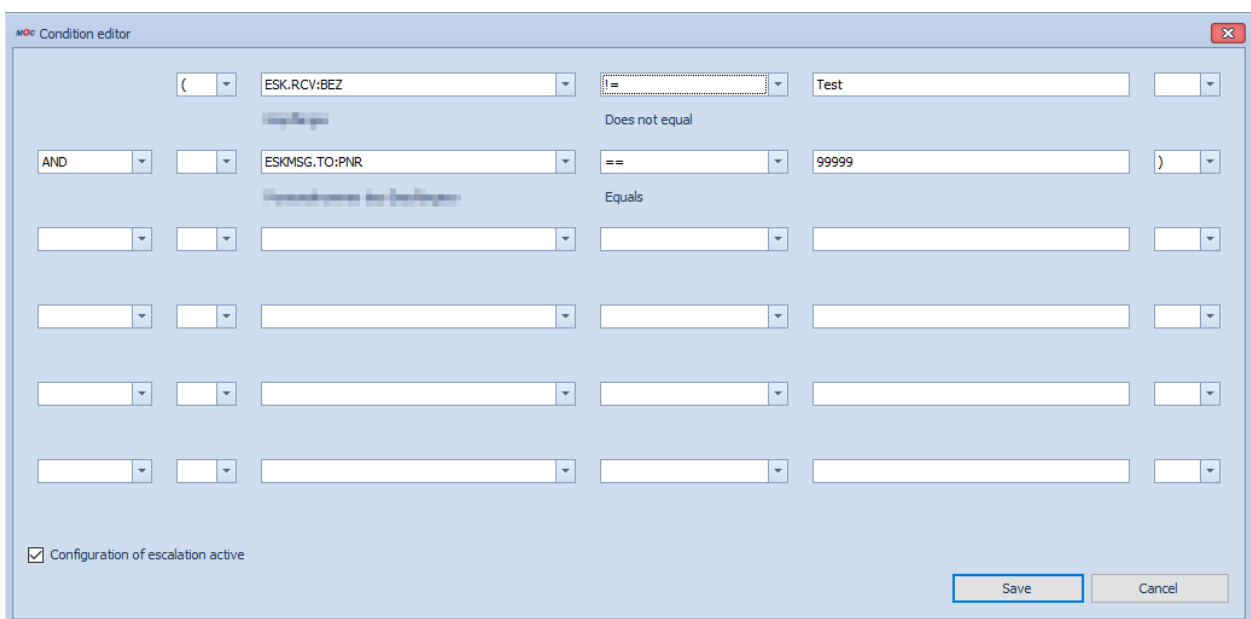
Condition editor

The conditions in the data records of the escalation configuration can become very complex. The *Condition editor* helps to edit the conditions and supports the user in the configuration of valid conditions.

The condition editor loads the condition of the data record currently selected. When the condition is edited and the condition editor is closed, then the condition is updated in the current data record.

If formula parsing is not possible, an error is displayed. If a formula stored in the system cannot be read when the condition editor is opened, an error is displayed.

Click the button *Condition editor* in the toolbar to call the condition editor. If a data record is selected, this data record is displayed in detail in the condition editor.



You can use the checkbox in the *Condition editor* to specify whether the escalation configuration is active or not.

Click the button *Save* to write the data to the current data record.



The condition editor does not know the data type of the variable and assumes that it is data type "string". If you use numeric variables with comparison operators with "greater" or "less", you must edit the created condition manually after using the editor. Delete the quotation marks. The comparisons will then work properly.

Tab *Last change*

Modified by

Last person who edited the escalation

Modified on

Date and time of the most recent modification